

Thematic Analysis of Capstone Research Posters: Assessing Scientific Reasoning Development in Genomics Training

Frederick Tan

C-MOOR / Clovis Community College Biology Department

Introduction

Understanding how undergraduate students develop into independent scientific researchers is a central challenge in genomics education. While quantitative metrics—such as the number of genes analyzed, databases used, or plots generated—can assess technical competence, they reveal little about the conceptual sophistication that distinguishes a novice from an emerging expert. Thematic analysis of the textual content of student capstone posters offers a window into this developmental process by examining the scientific reasoning, conceptual framing, and translational thinking that students employ when presenting their research. This report presents the results of a thematic analysis of twenty high-quality capstone research posters produced by college freshmen participating in the C-MOOR (Collaborative Microarray Open Online Research) genomics training program. The analysis focuses on five thematic dimensions: research motivation framing, disease relevance density, model organism justification, human ortholog integration, and pathway/network reasoning. Together, these dimensions

capture key aspects of scientific reasoning that are critical for translating molecular findings into biological insight—precisely the kind of higher-order thinking that genomics training programs aim to cultivate.

Methods

Twenty capstone research posters were collected from the Posters/high_quality directory, each consisting of a text transcript (*.txt) and a corresponding poster image (*.jpeg). The text transcripts were parsed and reviewed for all twenty posters. Each poster was independently coded across five thematic dimensions by reviewing the full text of each poster transcript, including the abstract, introduction, methods, results, and discussion sections. Coding was performed at the level of the entire poster rather than isolated sentences, to capture the primary intellectual framing of the research.

The five thematic dimensions were defined as follows. **Research motivation framing** classified each poster's primary motivation as either strongly disease-motivated (disease as the explicit central frame), translational (human relevance explicitly discussed), basic science/discovery (hypothesis-driven without disease framing), or applied/agricultural. **Disease mention density** catalogued all named human diseases mentioned anywhere in the poster and classified whether they appeared as a primary frame or as a secondary mention in the discussion. **Model organism justification** assessed the sophistication of the argument for why *Drosophila melanogaster* is an appropriate model for the research question, scored from minimal (generic lifecycle/ethics arguments) to sophisticated (specific pathway or mechanistic conservation arguments). **Human ortholog integration** evaluated whether and how deeply students incorporated human ortholog data—particularly tissue expression data from resources such as the Human Protein Atlas—into their analysis and reasoning. **Pathway/network thinking** assessed whether students situated their gene of interest within a broader

biological pathway, discussed multi-gene interactions, or engaged with systems-level reasoning, as opposed to treating the gene as an isolated finding.

All coding decisions were made by holistic review of each poster transcript, and the results are reported as frequency counts, categorical distributions, and illustrative examples. No automated text mining was employed; all analysis was performed by close reading.

Results

1. Research Motivation Framing

Each poster was classified by its primary stated motivation (Table 1).

Table 1. Research motivation framing across 20 posters.

Framing Category	Count	Percentage	Representative Poster
Basic science / discovery	6	30%	Amylase gene expression in Drosophila midgut (polysaccharide digestion compartmentalization)
Disease-motivated (primary)	7	35%	GALM and galactosemia; SLC45A2 and OCA4 albinism; Zellweger Spectrum Disorder; ASD and gut-microbiome-brain axis
Translational (human relevance explicit)	6	30%	CRAT as therapeutic target for metabolic disorders; fatty acid metabolism and

Framing Category	Count	Percentage	Representative Poster
			MCADD/cystic fibrosis; Toll-like receptor pathways and IBD
Applied / agricultural	1	5%	Cytochrome P450 and insecticide resistance

The dominant mode across the corpus is **disease-motivated or translational framing**, with 65% of posters explicitly connecting their gene of interest to human health applications. Only six posters (30%) maintained a purely discovery-science framing without anchoring to a disease or human application. This distribution suggests that the C-MOOR training is highly effective at helping students understand the **translational relevance** of molecular biology—even freshmen can articulate why a fly gene matters for human health.

2. Human Disease Mention Density

Table 2. Posters with named human diseases, by degree of centrality.

Category	Count	Posters
Disease as primary frame (abstract/introduction)	7	#3 (galactosemia), #7 (OCA4 albinism), #8 (Zellweger Spectrum Disorder), #11 (Autism Spectrum Disorder), #13 (colorectal cancer/KRAS), #15 (acid reflux/GERD)
Disease mentioned prominently in discussion	7	#2 (Zellweger's via paralogs), #4 (metabolic syndrome, obesity, fatty acid oxidation)

Category	Count	Posters
		disorders), #5 (MCADD, cystic fibrosis, COPD), #6 (rectum adenocarcinoma), #10 (Crohn's disease, IBD), #12 (Parkinson's disease), #14 (tumor suppression)
No named human disease	6	#1, #9, #16, #17, #18 (partial), #19, #20

Total: 14 of 20 posters (70%) mention at least one named human disease.

Disease categories represented across the corpus:

Disease Category	Specific Conditions	Posters
Metabolic / inherited metabolic	Galactosemia, MCADD, Zellweger Spectrum Disorder, fatty acid oxidation disorders	4
Gastrointestinal / digestive	Acid reflux/GERD, Crohn's disease, IBD, colorectal cancer	4
Neurological / developmental	Autism Spectrum Disorder, Parkinson's disease	2
Dermatological / pigmentation	Oculocutaneous albinism type 4	1
Oncology (general)	Tumor suppression, Ras-driven cancer	2

The predominance of gastrointestinal and metabolic diseases reflects the **gut-focused nature of the research context** (all posters analyzed gene expression in the *Drosophila* midgut).

Notably, students are not merely tacking on disease language as a rhetorical device—the disease framing is substantively connected to the biological question being investigated.

3. Model Organism Justification Sophistication

Table 3. Sophistication of Drosophila-to-human model organism justification.

Level	Description	Count	Posters
Sophisticated	Cites specific pathway/ortholog conservation with mechanistic argument	5	#4 (conserved metabolic pathways), #5 (fatty acid β -oxidation disorders), #8 (peroxisome biogenesis + 75% gene sharing + phenotypic parallels), #11 (gut-brain axis conservation), #13 (Ras signaling conservation in intestinal stem cells)
Intermediate	States human orthology, anatomical analogy, or “75% gene sharing”	11	#1, #2, #3, #6, #7, #9, #10, #12, #14, #15, #20
Minimal	Generic arguments (short lifecycle, cheap, ethical) only	4	#16, #17, #18, #19

The most sophisticated justification found was in Poster #8 (Zellweger Spectrum Disorder), which stated: “Approximately 75% of disease-causing genes in humans have a homolog in flies” and then enumerated specific phenotypic parallels (abnormal neuroanatomy, locomotor behavior, retinal issues, decreased lifespan in flies; vs. intellectual disability, vision problems, hypotonia, feeding difficulties in humans). This level of argument—connecting specific biological phenotypes across species—was rare; most students defaulted to generic orthology statements.

Growth target: Students would benefit from explicit instruction on how to argue for **mechanistic conservation** (why the pathway, not just the gene, is shared) rather than relying on the “75% of human disease genes are in flies” statistic alone.

4. Integration of Human Ortholog Analysis

Table 4. Depth of human ortholog data integration.

Level	Description	Count	Posters
Deep integration	Human tissue expression data used as core part of analysis; explicit cross-species comparison	4	#3 (GALM → Human Protein Atlas kidney expression), #6 (AMY2B → rectum adenocarcinoma prognosis), #11 (ASD genes → CNS + gut expression), #14 (FRK → human small intestine tumor suppressor)
Moderate	Human ortholog identified and function described;	8	#1, #2, #4, #5, #7, #8, #9, #12

Level	Description	Count	Posters
Minimal	used in discussion but not central	8	#10, #13, #15, #16, #17, #18, #19, #20
	Fly-only analysis; human ortholog mentioned but not actively used		

Poster #3 (GALM/galactosemia) exemplifies deep integration: after finding that Galm1 is most highly expressed in the p2_4 region of the fly midgut (specializing in hexose metabolism), the student confirmed that the human ortholog GALM is also expressed most highly in human kidney tissue—directly connecting the fly regional specialization to the human tissue most susceptible to galactosemia pathology.

Poster #11 (ASD and the gut-microbiome-brain axis) also stands out for using the Human Protein Atlas to confirm that autism-related genes are expressed in both the central nervous system and the gastrointestinal tract in humans, providing the biological basis for the gut-brain axis hypothesis.

5. Pathway and Network Thinking

Table 5. Extent of pathway and systems-level reasoning.

Level	Description	Count	Posters
Systems-level	Gene situated in named pathway; multi-gene or paralog family interactions discussed; cellular compartment context	5	#2 (ScpX paralogs, peroxisome cluster, KEGG pathway), #3 (Leloir pathway of galactose metabolism), #5 (fatty acid β -oxidation, peroxisome,

Level	Description	Count	Posters
Moderate	Pathway named but single-gene focus; or pathway context acknowledged without multi-gene analysis	7	mitochondria), #8 (peroxisome biogenesis, lipid metabolism), #13 (Ras signaling pathway, tumor suppression network)
			#1 (starch/sucrose metabolic pathway), #4 (TCA cycle, acetyl-CoA), #9 (TALE gene developmental hierarchy), #10 (immune pathway categories listed separately), #16 (kynurenine pathway), #18 (CYP3/CYP4 clans), #20 (tyrosine kinase signaling)
Minimal / isolated	Gene treated as isolated finding; no pathway context	8	#6, #7, #11, #12, #14,

Level	Description	Count	Posters
			#15, #17, #19

Posters #2 (ScpX paralogs) and #5 (fatty acid metabolism) represent the highest level of systems thinking in the corpus. Poster #2 tracked not just ScpX but its three paralogs (CG17597, CG11151, euc, Hsdl2), mapped them to the peroxisome cluster via clusterProfiler, identified which paralogs share cholesterol transport function and which do not, and explicitly framed the research around **gene family functional diversification**—a genuinely sophisticated systems biology argument. Poster #5 similarly mapped three genes (Acox3, Muc68e, Mdh1) to distinct cellular compartments and pathways (peroxisome β -oxidation, mucus barrier, TCA cycle) and discussed how they cooperate in gut homeostasis.

The largest growth gap identified is in this dimension: 40% of posters treat genes as isolated findings, and even among those who name pathways, most do not analyze how multiple genes interact within them.

Discussion and Recommendations

The thematic analysis reveals several distinct patterns in how students frame and reason about their genomics research. The most striking finding is the **strong translational orientation** of the student cohort: 65% of posters explicitly frame research questions in terms of human disease or clinical relevance, and 70% mention at least one named human disease. This suggests the training is highly successful at helping students understand the **why** of their research—not just what the gene does in the fly, but what it means for human health.

However, several critical areas for student growth are apparent from this analysis:

1. **Pathway and network reasoning is the most underdeveloped skill.** Forty percent of posters treat genes as isolated findings, and even students who name pathways typically focus on single genes rather than multi-gene interactions. Helping students think of genes as nodes in networks rather than solo actors is a central challenge in modern biology education and represents the clearest growth target.
2. **Human ortholog integration is underutilized.** Only 20% of posters deeply integrate human tissue expression data from resources like the Human Protein Atlas. The majority mention an ortholog but do not use it as part of the analytical argument. Incorporating human expression data systematically would strengthen students' ability to make cross-species claims.
3. **Model organism justifications rarely move beyond generic orthology.** While most students can state that flies share genes with humans, few can articulate why a specific pathway is mechanistically conserved. Teaching students to argue for pathway-level rather than gene-level conservation would strengthen their scientific reasoning.

These findings align with the broader educational research literature suggesting that systems-level thinking and translational reasoning are higher-order skills that develop more slowly than technical proficiency. The data suggest that targeted interventions—perhaps structured exercises in pathway mapping (e.g., KEGG pathway exploration) and cross-species expression comparison—could accelerate growth in these dimensions.

Appendix: Poster Identification

Poster ID	File Name	Gene(s) Studied
1	1cf8d649fafc9dbf3dd99dbc89ac0439e657b02d	amyrel, mal-A1, mal-A8, mal-A4, hex-C, treh
2	24d4c6e39c72dabfb6818b4040bfd6bb071c2fe5	ScpX, CG17597, GC11151, euc, Hsdl2
3	2642603622963a937128325fb548a6b5c1b6e3e6	Galm1 / GALM
4	2bf8e85c882ad2cf618eb01100c56e0eae36dee5	CRAT, MANF, NPY
5	467feb559f59080b9e9578534dfca99394a57dd5	Acox3, Muc68e, Mdh1
6	4fd568791ed2cfa616767ae36c15bd99fada4e26	Amy-p / AMY2B
7	595a8d9fb070c3f922e4c9947eaa79813b783a55	SLC45A2
8	5a4835d6b146d6d8c2c810845002798107ab1dba	ABCD1, ACBD5, ACOX1, DNM1L, HSD17B4, SCP2
9	7ee7fbaccb55a104d3f69386918015c2dedf9144	TALE homeobox genes
10	9a0db552697e7306096dd2b271bc6884e881253c	Toll, JAK/ STAT, IMD, Spatzle
11	9abde11bd7c123eb6eeee36cd2e8432fb3cf75cf	Autism spectrum

Poster ID	File Name	Gene(s) Studied
		disorder candidate genes
12	a952ca5819a03af3bba3cbb0ca101222cd02170d	Yellow gene family
13	bb66c49ad7b64a2820f79e650a62b59b7a64bbef	Ras85D
14	d1cf18b2245419b48fe18b43c5c40e6669b5e3ea	Src42A / FRK
15	dab9ebfe20ae469ac0646a71f990099efaa60b38	Vha16-1 / ATP6V0C
16	e085d9c71b41d72ab14b1a3c892723b88be7cebb	white, vermillion, Speck, Ddc
17	e2cdaf53c1e79a229746cd526018bf425490577c	V-ATPase subunits
18	fbed8d367c3f418e925a5603c43ebb41d57bd53f	Cytochrome P450 (Cyp3/Cyp4 clans)
19	(partial coverage)	Tryptophan-kynurenine pathway genes
20	(partial coverage)	Src42A / FRK (tyrosine kinase)

C-MOOR is supported in part by NSF DUE Award #2021013 to Carnegie Institution, Johns Hopkins University, and Clovis Community College.